

Multiplying a vector by a number; the coordinates of a vector

Stretching, shrinking and reversing — and the test for parallel that comes free with it.

LESSON 42–43

2 × 40 MIN

GEOMETRY 8, ТЕМЫ 38–39

STAGE 9 · 13.2

LESSON CLOCK

6'

12'

12'

6'

4'

Double a vector on squared paper	6 min
Multiplying by a number	12 min
Coordinates of a vector	12 min
The collinearity test	6 min
Homework	4 min

LEARNING OBJECTIVES

- Multiply a vector by a scalar, and describe the effect on length and direction.
- Write a vector in coordinates from the coordinates of its ends.
- Test two vectors for collinearity.
- Use vectors to prove that a line is parallel to another.

TEXTBOOK

National	Темы 38–39, pp. 85–89
Cambridge	Learner’s Book pp. 291–296
Workbook	Workbook 13.2

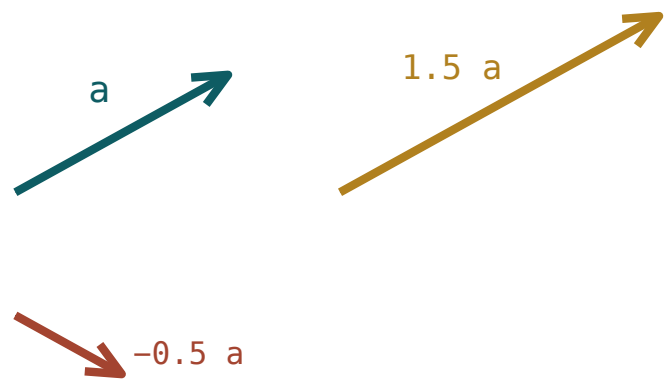
01

Explanation

Multiplying by a number

SCALAR MULTIPLE

$k \cdot (a_1; a_2) = (ka_1; ka_2)$ and $|ka| = |k| \cdot |a|$



a number stretches or reverses a vector

*Multiplying by a number stretches or shrinks the arrow;
a negative number also turns it round.*

<i>K</i>	EFFECT
$k > 1$	longer, same direction
$0 < k < 1$	shorter, same direction
$k = 0$	the zero vector
$k < 0$	direction reversed, length $ k $ times

Note the modulus signs: $|-3\mathbf{a}| = 3|\mathbf{a}|$. A length is never negative, however negative the multiplier.

Coordinates, and the test for parallel

The coordinates of a vector are found once and used everywhere:

$$AB = (x_B - x_A; y_B - y_A)$$

“End minus start” — and if you swap them you get the opposite vector, which is the commonest slip in this chapter.

TEST FOR COLLINEAR VECTORS

$\mathbf{a}$ and $\mathbf{b}$ are collinear exactly when $\mathbf{b} = k\mathbf{a}$ for some number k — that is, when

$$a_1b_2 - a_2b_1 = 0$$

Check $(2; 3)$ and $(6; 9)$: $2 \cdot 9 - 3 \cdot 6 = 0$ ✓, and indeed $(6; 9) = 3(2; 3)$. Check $(2; 3)$ and $(6; 8)$: $2 \cdot 8 - 3 \cdot 6 = -2 \neq 0$ — not collinear.

PARALLEL LINES FROM PARALLEL VECTORS

If AB and CD are collinear vectors and the four points are not all on one line, then the lines AB and CD are parallel. That is how vectors prove parallelism — no angles, no measuring.

02 Worked examples

EXAMPLE 1 $\mathbf{a} = (-2; 5)$. Find $3\mathbf{a}$, $-\mathbf{a}$ and $|\mathbf{3a}|$.

① $3\mathbf{a} = (-6; 15)$

Multiply each coordinate.

② $-\mathbf{a} = (2; -5)$

Reverse both signs.

③ $|\mathbf{a}| = \sqrt{4 + 25} = \sqrt{29}$

④ $|\mathbf{3a}| = 3\sqrt{29}$

Three times as long.

ANSWER

$$(-6; 15), (2; -5), 3\sqrt{29}$$

EXAMPLE 2 $A(1; 2), B(4; 6), C(0; 1), D(6; 9)$. Is $AB \parallel CD$?

① $AB = (3; 4)$

End minus start.

② $CD = (6; 8)$

③ $3 \cdot 8 - 4 \cdot 6 = 0$

The collinearity test.

④ $CD = 2 \cdot AB$

So the vectors are parallel — and C is not on line AB .

ANSWER

Yes, $AB \parallel CD$ and CD is twice as long

EXAMPLE 3 Find a unit vector in the direction of $(9; 12)$.

① $|(9; 12)| = \sqrt{81 + 144} = 15$

② $\frac{1}{15} \cdot (9; 12)$

Divide by the length.

③ $(0.6; 0.8)$

④ $\sqrt{0.36 + 0.64} = 1$

Check: the length really is 1.

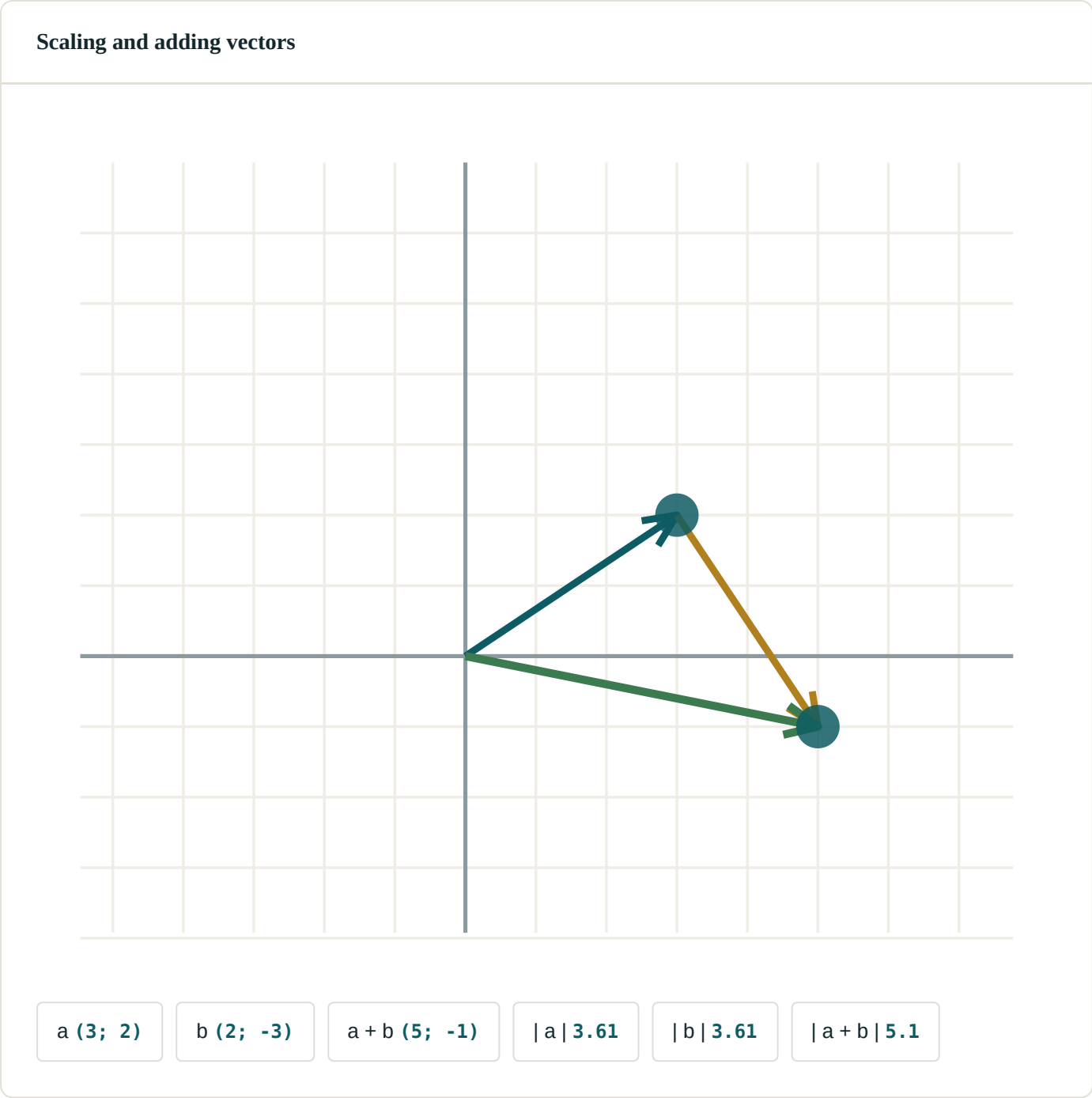
ANSWER

$(0.6; 0.8)$

03

Interactive model

Drag a tip and watch how each coordinate scales when the vector is stretched.



04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Scalar multiple	Skalyarga ko'paytma	Умножение на число
Stretch	Cho'zish	Растяжение
Compression	Siqish	Сжатие
Direction reversed	Yo'nalish teskari	Направление противоположное
Coordinates of a vector	Vektor koordinatalari	Координаты вектора
Collinear	Kollinear	Коллинеарные
Unit vector	Birlik vektor	Единичный вектор
Position vector	Radius-vektor	Радиус-вектор
Parallel	Parallel	Параллельный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$4 \cdot (2; -3)$ is:

$(8; -12)$

$(6; 1)$

$(8; 12)$

$(2; -12)$

If $|a| = 6$ then $|-2a|$ is:

-12

12

4

3

$(3; 5)$ and $(9; 15)$ are:

perpendicular

collinear

equal

opposite

For $A(2; 3)$ and $B(5; 1)$, AB is:

$(3; -2)$

$(-3; 2)$

$(7; 4)$

$(3; 2)$

A unit vector has length:

0

1

any

10

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find $2 \cdot (3; 4)$.

ANSWER $(6; 8)$

2. Find $-1 \cdot (5; -2)$.

ANSWER $(-5; 2)$

3. Find $0.5 \cdot (8; 6)$.

ANSWER $(4; 3)$

4. If $|a| = 4$, find $|3a|$.

ANSWER 12

5. Find AB for $A(0; 0)$, $B(-3; 7)$.

ANSWER $(-3; 7)$

6. Are $(1; 2)$ and $(2; 4)$ collinear?

ANSWER Yes.

7. Find $|2 \cdot (3; 4)|$.

ANSWER 10

Medium · the standard question

7 PROBLEMS

1. Find $3 \cdot (-2; 5) - 2 \cdot (1; 4)$.

ANSWER $(-8; 7)$

2. If $|a| = 5$, find $|-4a|$.

ANSWER 20

3. Are $(4; -6)$ and $(-6; 9)$ collinear?

ANSWER Yes — $4 \cdot 9 - (-6)(-6) = 0$

4. Are $(2; 5)$ and $(4; 9)$ collinear?

ANSWER No — $2 \cdot 9 - 5 \cdot 4 = -2 \neq 0$

5. $A(1; 1)$, $B(3; 5)$, $C(2; 0)$, $D(6; 8)$. Is $AB \parallel CD$?

ANSWER Yes — $CD = 2 \cdot AB$

6. Find a unit vector in the direction of $(3; 4)$.

ANSWER $(0.6; 0.8)$

7. Find k so that $(6; k)$ is collinear with $(2; 5)$.

ANSWER $k = 15$

Hard · stretch and reason

7 PROBLEMS

1. $a = (2; -1)$, $b = (-3; 4)$. Find $2a - 3b$ and its length.

ANSWER $(13; -14)$, length $\sqrt{365}$

2. Find all k for which $(k; 6)$ and $(3; k)$ are collinear.

ANSWER $k^2 = 18$, $k = \pm 3\sqrt{2}$

3. Find a vector of length 20 in the direction of $(-3; 4)$.

ANSWER $(-12; 16)$

4. $A(0;0)$, $B(6;4)$. Find the point P on AB with $AP = \frac{2}{3} \cdot AB$.

ANSWER $P(4; \frac{8}{3})$

5. M and N are midpoints of AB and AC . Show that $MN \parallel BC$ and $MN = \frac{1}{2}BC$, using vectors.

ANSWER $MN = AN - AM = \frac{1}{2}AC - \frac{1}{2}AB = \frac{1}{2}BC$ — a multiple of BC , so parallel and half as long

6. $a = (1; 2)$ and $b = (3; k)$. Find k so that a and b are collinear, and then find b in terms of a .

ANSWER $k = 6$, $b = 3a$

7. Explain why the collinearity test $a_1b_2 - a_2b_1 = 0$ works.

ANSWER If $b = ka$ then $a_1(ka_2) - a_2(ka_1) = 0$; conversely, when the expression is zero the ratios $\frac{b_1}{a_1}$ and $\frac{b_2}{a_2}$ agree, giving a common multiplier k .

07 Homework

Homework — 6 problems

Geometry 8, Темы 38–39, pp. 85–89. Show the collinearity test in full.

- Find $5 \cdot (-1; 3)$ and its length.
- If $|a| = 7$, find $|-3a|$.

3. Are $(6; -4)$ and $(-9; 6)$ collinear? Justify.
4. Find k so that $(4; k)$ is collinear with $(3; 12)$.
5. Find a unit vector in the direction of $(5; 12)$.
6. $A(1;0)$, $B(4;6)$, $C(0;2)$, $D(2;6)$. Is $AB \parallel CD$?